

AI/ML Internship — Day 1 Report

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What is Pose Detection?

Introduction

Pose detection is a computer vision and Artificial Intelligence (AI) technology used to identify and track human body landmarks from images or videos. It helps computers understand body posture, movement, and gestures by detecting different body parts such as the head, shoulders, elbows, wrists, hips, knees, and ankles.

Pose detection systems are widely used in modern AI applications because they allow machines to visually understand human movement in real time. This technology is an important part of computer vision and human-computer interaction systems.

How Does Pose Detection Work?

Pose detection generally works in the following steps:

1. Capture of Image or Video

A camera or webcam captures images or live video frames of a person.

2. Human Body Detection

The AI model identifies the presence of a human body in the image or video frame.

3. Detection of Body Key Points

Important body landmarks such as:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

are detected by the AI system.

4. Formation of Digital Skeleton

The detected body points are connected to form a digital body skeleton structure.

5. Movement and Posture Analysis

The AI system analyses body posture, movement, and gestures in real time.

Technologies Used

Common technologies used in pose detection systems include:

- Python
- OpenCV
- MediaPipe
- TensorFlow
- Deep Learning Models

In this internship task, OpenCV and MediaPipe were used to implement real-time pose detection using a webcam.

Applications of Pose Detection

Pose detection is used in many industries and applications, including:

- fitness and exercise tracking
 - yoga posture correction
 - sports performance analysis
 - gaming and motion sensing
 - physiotherapy and rehabilitation
 - healthcare monitoring
 - virtual try-on systems
 - gesture recognition
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Benefits of Pose Detection

Advantages of pose detection include:

- real-time human body tracking
 - accurate movement analysis
 - improved human-computer interaction
 - no requirement for physical sensors
 - wide range of AI applications
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Conclusion

Pose detection is an important AI and computer vision technology that helps computers understand human posture and movement. It is widely used in fitness, healthcare, gaming, sports, and fashion industries. With advancements in AI technologies, pose detection systems are becoming more accurate, faster, and more efficient in real-world applications.

How Can AI Help Fashion Shopping?

Introduction

Artificial Intelligence (AI) is transforming the fashion industry by making shopping experiences more personalized, intelligent, and interactive. AI technologies help customers choose products more easily while helping companies improve customer satisfaction and reduce product returns.

Modern online shopping platforms use AI to analyse customer behaviour, recommend products, estimate clothing sizes, and provide virtual try-on experiences.

Applications of AI in Fashion Shopping

1. Virtual Try-On Technology

Virtual try-on systems allow customers to digitally try clothes, shoes, glasses, or accessories using AI and augmented reality technologies.

The AI system detects the customer's body position and overlays virtual products on the user in real time. This improves customer confidence before purchasing products online.

2. AI-Based Body Measurement

AI systems can estimate body measurements using images or videos.

Measurements estimated include:

- height
- waist size
- shoulder width
- chest size

This helps customers choose accurate clothing sizes and reduces incorrect purchases.

3. Personalized Recommendations

AI analyses:

- customer preferences
- browsing history
- shopping behaviour
- fashion interests

Based on this analysis, AI recommends suitable clothing and fashion products.

4. Fashion Trend Analysis

AI helps fashion companies analyse:

- market trends
- customer interests
- social media data
- seasonal demand

This helps businesses predict fashion trends and manage inventory more efficiently.

5. Visual Search Technology

AI-powered visual search systems allow users to search products using images instead of text descriptions.

For example:

- uploading a dress image to find similar products online
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Technologies Used

AI fashion systems commonly use:

- Artificial Intelligence
- Machine Learning
- Computer Vision
- Deep Learning
- Pose Detection
- Augmented Reality

Benefits of AI in Fashion Shopping

Advantages of AI in fashion shopping include:

- improved customer experience
- personalized shopping
- accurate size prediction
- reduced product returns
- faster product discovery
- better fashion recommendations

Internship Relevance

In this internship task, concepts related to:

- pose detection
- body tracking
- computer vision
- virtual try-on systems

were explored to understand how AI technologies are applied in modern fashion applications.

Conclusion

AI is playing a major role in improving the fashion shopping experience by making online shopping smarter, faster, and more personalized. Technologies such as virtual try-on systems, AI-based body measurements, and recommendation systems are helping customers make better purchasing decisions while improving efficiency for fashion companies.
